

Infinite-Dimensional Diagonals in Noncommutative Function Theory: An Explicit Later Answer

Literature status for arXiv:math/0603437

13 August 2026

Status: literature already answered. Ueda’s arXiv:0802.3449 explicitly removes the finite-dimensionality assumption on the diagonal from exactly the source’s Corollary 3.5, Theorems 4.1–4.2, and Corollary 4.3. This includes both noncommutative Gleason–Whitney statements and the Kaplansky-density conclusion.

The source question

Let A be a maximal subdiagonal algebra in a finite von Neumann algebra M with faithful normal tracial state, and let $D = A \cap A^*$ be its diagonal. Under the assumption $\dim D < \infty$, the source proves the following results:

- Corollary 3.5: a functional annihilating $A + A^*$ has zero normal part;
- Theorem 4.1: normal functionals on A have both relevant Gleason–Whitney conclusions;
- Theorem 4.2: each $\omega \in M_*$ is the unique Hahn–Banach extension of $\omega|_{A+A^*}$;
- Corollary 4.3: the unit ball of $A + A^*$ is weak* dense in the unit ball of M .

Immediately after Corollary 4.3, the source says that it does not know whether “the last few results” hold without finite-dimensionality of D . This appears on source PDF page 12 [1].

Ueda’s exact answer

Ueda proves a peak-set theorem for every noncommutative H^∞ -algebra $A = H^\infty(M, \tau)$, with no finite-dimensional restriction on D . Theorem 4.1 of the supporting paper says that whenever a bounded functional φ annihilates A , its normal and singular parts separately annihilate A . Ueda then observes that the original proofs need only this weaker F. and M. Riesz consequence, rather than the stronger finite-dimensional theorem used in the source.

The paragraph immediately before Ueda’s next theorem states that the proofs of the source’s Corollary 3.5, Theorems 4.1–4.2, and Corollary 4.3 therefore all work without any dimensional assumption. Ueda’s theorem on supporting PDF pages 6–7 records the resulting four conclusions explicitly:

1. a functional annihilating $A + A^*$ is singular;

2. every Hahn–Banach extension to M of a normal functional on A is normal;
3. every $\varphi \in M_*$ is the unique Hahn–Banach extension of $\varphi|_{A+A^*}$, with preservation of norm;
4. every element of M is weak* approximable by a norm-bounded net from $A + A^*$.

Thus the finite-dimensional-diagonal question in the source has an unqualified affirmative answer [2].

Independent author confirmation

The source authors later make the provenance completely explicit. In the introduction of arXiv:1605.08932 they state that Ueda removed the dimensional restriction from “four or five” of their theorems. Their Theorem 3.2 on supporting PDF page 8 again proves, for every maximal subdiagonal algebra in a σ -finite von Neumann algebra,

$$\text{Ball}(A + A^*) \text{ is weak* dense in } \text{Ball}(M),$$

and explicitly identifies this as the removal of the restriction in the source’s Corollary 4.3. Corollary 5.9 on supporting PDF page 17 gives the corresponding Gleason–Whitney equivalences and says that they all hold for maximal subdiagonal algebras [3].

The original source setting already assumes a faithful normal tracial state, so M is σ -finite. Consequently, the later confirmation applies directly and also extends beyond the tracial setting.

Scope and remaining question

This packet settles only the source remark about removing $\dim D < \infty$. It does not settle the separate Corollary 2.6 question asking whether the unique normal state extension property alone forces a tracial algebra to be maximal subdiagonal. A separate attempt note records a focused investigation of that surviving problem.

Verification and provenance

The source remark was checked on source PDF page 12. Ueda’s exact list was checked in Section 4 on supporting PDF pages 6–7. The source authors’ confirmation was checked in the introduction and in Theorem 3.2 and Corollary 5.9 of arXiv:1605.08932 (PDF pages 1–2, 8, and 17). Ueda cites the source and identifies each result by its original number, so this is an explicit literature answer rather than an inferred theorem match.

References

- [1] D. P. Blecher and L. E. Labuschagne, *Noncommutative function theory and unique extensions*, Studia Math. **178** (2007), 177–195; arXiv:math/0603437.
- [2] Y. Ueda, *On peak phenomena for non-commutative H^∞* , Math. Ann. **343** (2009), 421–429; arXiv:0802.3449.
- [3] D. P. Blecher and L. E. Labuschagne, *Ueda’s peak set theorem for general von Neumann algebras*, Trans. Amer. Math. Soc. **370** (2018), 8215–8236; arXiv:1605.08932.